Inertia Governors and Stability of Governor

by

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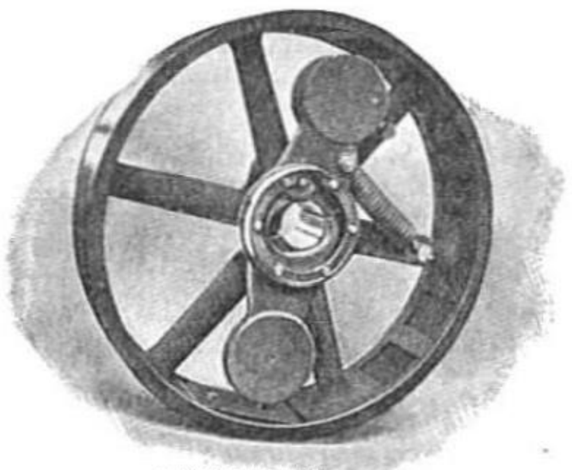
Inertia Controlled Governors :

- Inertia governors works on the principle of movement of inertia due to both centrifugal and inertia force.
- These governors: Not being used frequently.
- Inertia governors are more sensitive than the centrifugal governors, and it becomes difficult to completely balance the revolving parts.

Westinghouse inertia governor



GOVERNORS by Rajesh Kumar
www.echeloninstitute.org

Fig. 1

Sensitiveness of Governor:

- Consider **two governors A and B running at the same speed.** When their speed increases or decreases by a certain amount, the lift of the sleeve of governor A is greater than the lift of the sleeve of governor B. **Therefore, the governor A is more sensitive than the governor B.**

- It is a ratio of the **difference between the maximum and minimum equilibrium speed to the mean equilibrium speed.**

- **Sensitiveness of governor = $(N_2 - N_1) / N$**; Where N = (N1 + N 2) / 2 (1)

- **Sensitiveness of governor** = $\dfrac{2(\omega_2 - \omega_1)}{\omega_1 + \omega_2}$ (2)

Here, N_1 = minimum equilibrium speed and N_2 = maximum equilibrium speed

- The **sensitiveness of the governor increases as the speed range decreases**.

Stability of Governor:

- **Ideal case-** Radius of rotation of governor constant, at every speed.
- **What happened practically? Speed increases with radius increases** and **Speed decreases with radius decreases.**
- **A governor is said to be stable** when for every speed with in the working range there is a definite configuration i.e. **there is only one radius of rotation of the fly balls**.
- **For a stable governor: if speed increases, the radius of governor ball must be increase.**
- **For a unstable governor: if speed increases, the radius rotation decrease.**

Isochronous Governors:

A governor is said to be ***isochronous*** **when the equilibrium speed is constant for all radii of rotation of the balls within the working range, neglecting friction.**

- **The range of speed is zero, if N_2 - N_1 = 0 than N_2 = N_1**

Case 1: Porter governor:

$$N_1^2 = \left[\frac{m+M}{m}\right]\frac{895}{h_1} \quad (3)$$

$$N_2^2 = \left[\frac{m+M}{m}\right]\frac{895}{h_2} \quad (4)$$

For isochronism, range of speed should be zero *for example* **N_2 - N_1 = 0 than N_2 = N_1** Therefore from equations **(*3*) and (*4*), *h1 = h2*, which is impossible in case of a Porter governor.** Hence a ***Porter governor cannot be isochronous***.

Case 2: Hartnell governor

$$M.g + S_1 = 2F_{C1} \times \frac{x}{y} = 2m \times \left(\frac{2\pi N_1}{60}\right)^2 \times r_1 \times \frac{x}{y}$$ (5) Minimum speed

$$M.g + S_2 = 2F_{C2} \times \frac{x}{y} = 2m \times \left(\frac{2\pi N_2}{60}\right)^2 \times r_2 \times \frac{x}{y}$$ (6) Maximum speed

If $N_1 = N_2$, from equation (5) and (6)

$$\frac{M.g + S_1}{M.g + S_2} = \frac{r_1}{r_2}$$ (7)

Note : The isochronous governor is not of **practical use because the sleeve will move to one of its extreme positions immediately the speed deviates from the isochronous speed.**

Hunting of Governor:

A governor is said to be hunt if speed of the engine fluctuates continuously above and below the mean speed.

This is caused by too sensitive governor which changes the fuel supply by a large amount when a small changes in speed of rotation takes place.

Effort of Governor:

The effort of governor is the mean force acting on the sleeve to raise or lower it for a given change of speed.

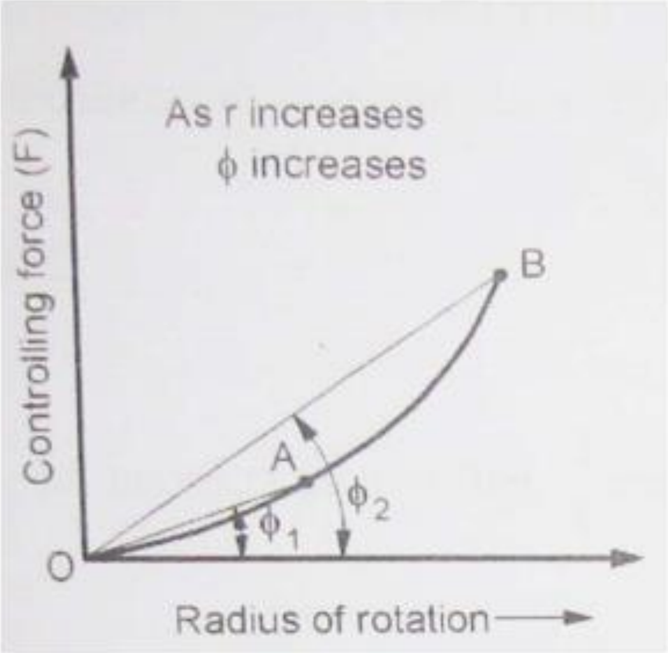
Power of a governor:

Power of governor is the work done at the sleeve for a given percentage change of speed. It is the product of the mean value of the effort and the distance through which the sleeve moves.

Power = Mean effort × distance moved by sleeve

(A) Stable governor

For stable governor as radius of rotation increase the controlling force must increase. Means, **Fc/r** must increase as ***r*** increases. Therefore controlling force curve DE must intersect the controlling force axis below origin. The equation of curve DE becomes **Fc = Ar - B**



Stable governor:
As 'r' increase 'angle ' must increase.

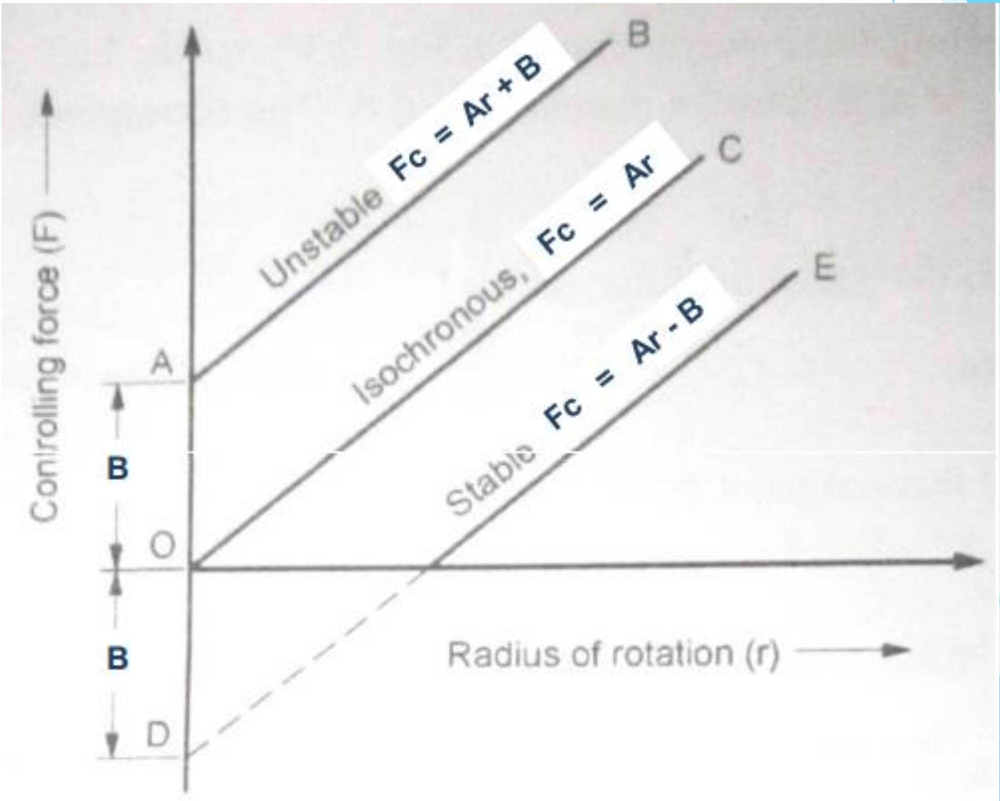
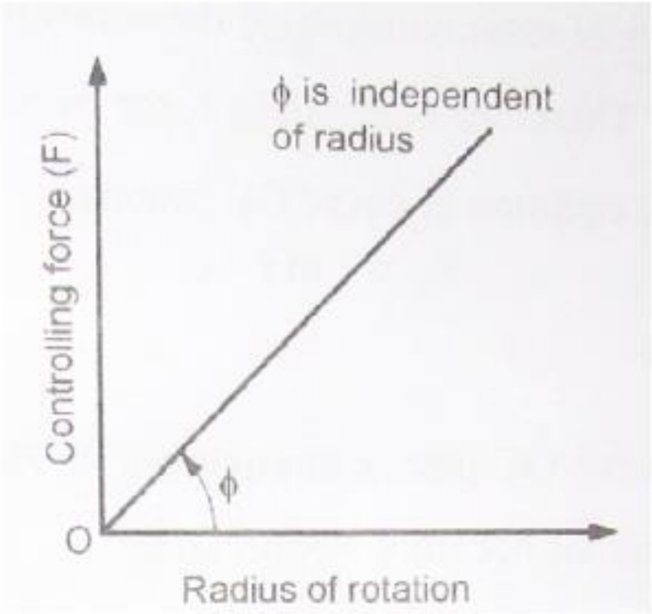


Fig. 2

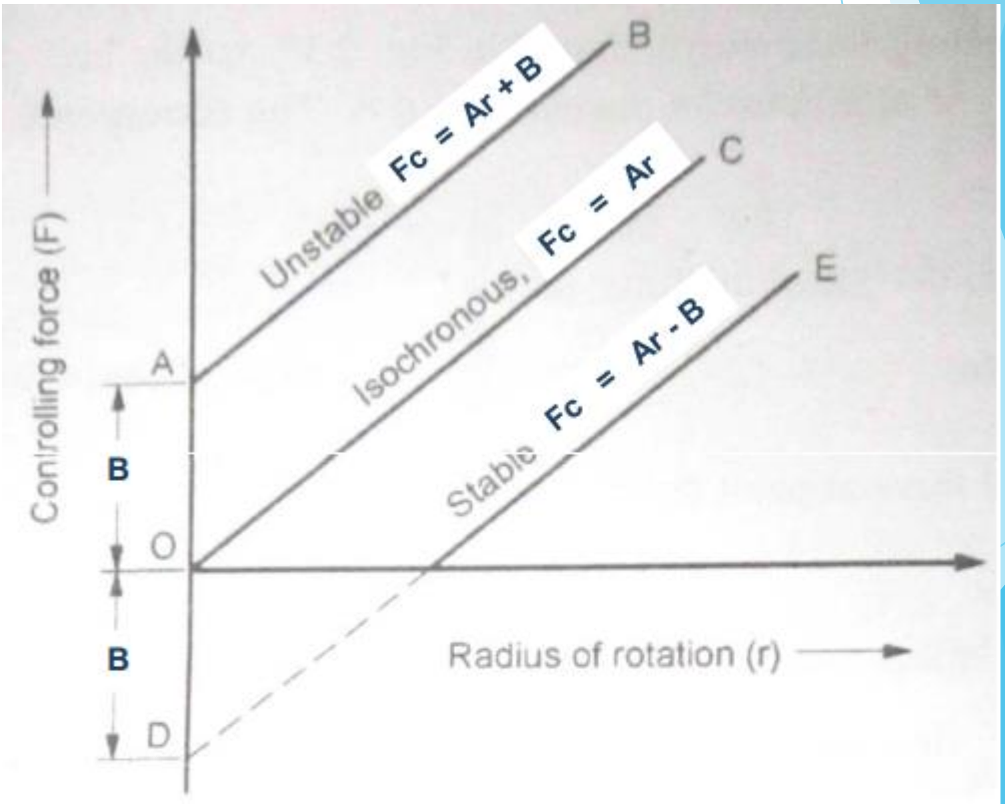
Controlling force diagram for spring loaded governors

(B) Isochronous governor:

It B = 0 controlling force curve OC passes through origin. Hence Fc/r will remain constant for radius of rotation. Thus governor becomes isochronous. Equation of line OC is Fc = Ar



Isochronous governor: 'angle' is constant and is independent of radius

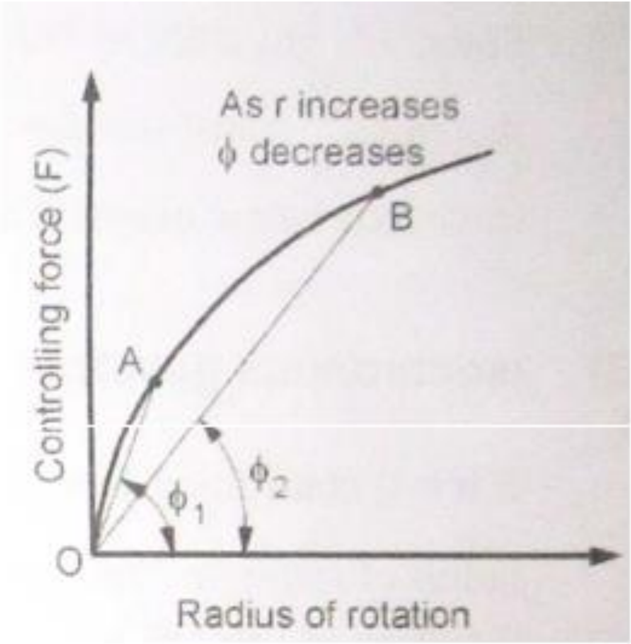


Controlling force diagram for spring loaded governors

Fig. 3

(C) Unstable governor:

If B is positive, then Fc/r decreases as r increases. It means that as the radius of governor increases the controlling force decreases. Thus governor becomes unstable Equation of line AB is Fc = Ar+ B



Unstable governor:
As 'r' increases 'angle ' will decrease.

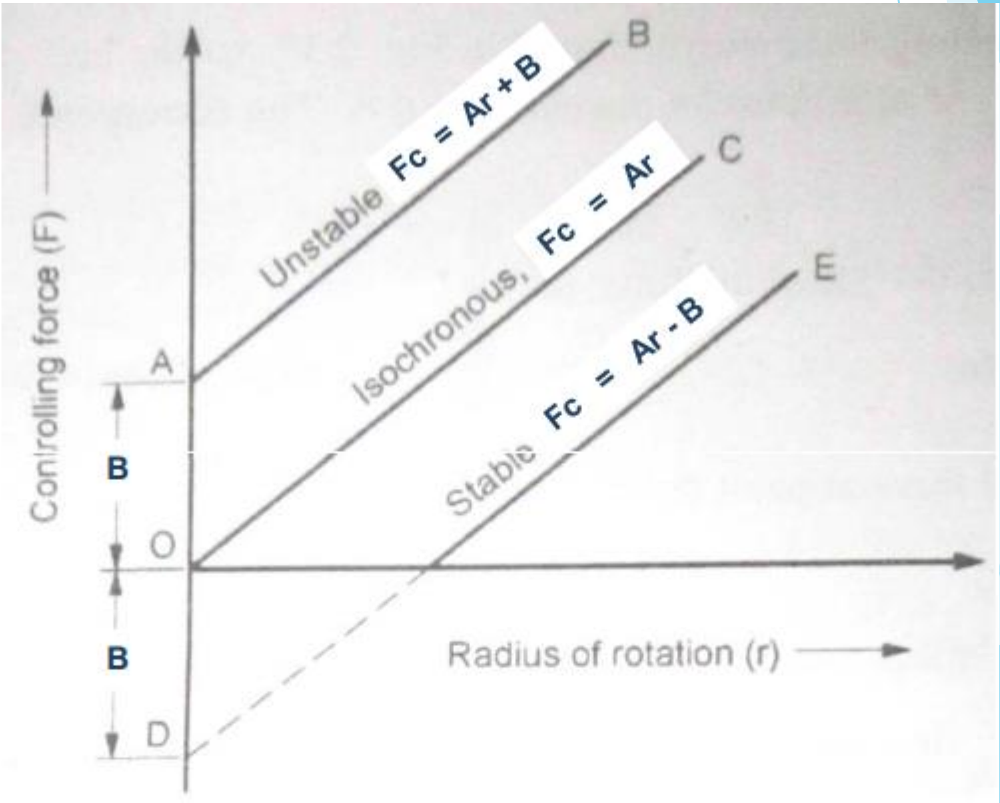


Fig. 4

Controlling force diagram for spring loaded governors

Problem 1. In the case of unstable governor, explain clearly what will happen when the speed of rotation is first increased from zero to the maximum equilibrium speed and is then diminished below the minimum equilibrium speed.

- At this situation of governor: it is the highest degree of hunting.

- When the governor is at rest, the balls are at their minimum distances from the axis of rotation and the sleeve is in its lowest position.
- As the speed of rotation increases no movement of the ball or sleeve takes place until the equilibrium speed which corresponding to the minimum radius of rotation is reached.
- The slightest further increase of speed then upsets the equilibrium and immediately causes the governor ball to fly out to their maximum radius.

- This has the effect of cutting off the supply of energy to the engine and the speed begins to fall.
- The speed continues to fall until the equilibrium speed which corresponds to the maximum radius of rotation is reached.
- The slightest further decrease of speed once more disturbs the equilibrium and the balls immediately return to their minimum radius.

Thank You